

BILL BOEING

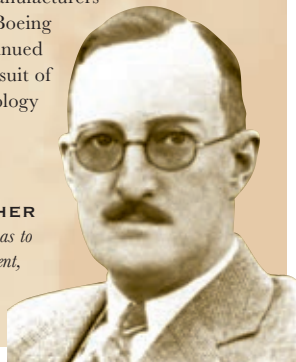
BORN IN DETROIT, WILLIAM E. BOEING

(1881–1956) studied engineering at Yale before following his father into the lumber business. In 1914 he bought himself a seaplane, but decided he could build a better one himself. With his friend US Navy Commander Conrad Westervelt, Boeing created the B&W seaplane and, in 1916, set up a company to manufacture it – Pacific Aero Products in Seattle. Relying on his timber and furniture business to keep him going during hard times, Boeing expanded his aviation interests throughout the 1920s, not only building airplanes but also creating the Boeing Air Transport Company to carry US airmail and passengers. The conglomerate of aircraft manufacture, air transport, and related companies that Boeing put together became an obvious

target for New Deal trust-busters after President F.D. Roosevelt's election in 1933. Boeing became heatedly involved in political disputes and retired in 1934 in protest at the decision to forcibly separate aircraft manufacturers from airlines. The Boeing Company has continued to combine the pursuit of cutting-edge technology with acute business acumen.

FOUNDING FATHER

Boeing founded what was to become, after his retirement, the world's leading aircraft business.



herself was regarded as an obstacle. The priorities on early 1930s flights were, from top to bottom: mail, passengers, stewardesses. If the pilot decided his airplane was overweight for takeoff, he would dump the stewardess and continue without her.

From 1930, travelers tired of the discomforts of Fokker, Ford, and Boeing trimotors found relief on some routes in the Curtiss Condor. Flown first by Eastern and then by American Airways, it was

a design that, like most British interwar airliners, sacrificed looks and performance in the cause of passenger comfort. An old-fashioned strut-and-fabric biplane with a tendency to ice up and engines disturbingly prone to catch fire, the Condor could hardly be termed a technical success. Yet its wide fuselage offered a new level of luxury, including 12 sleeping berths for use on overnight journeys. In publicity for the Condor, much was made of the fact that a radio set installed in the passenger cabin relayed music and news during the flight. The selling point was not so much the in-flight entertainment as the fact that passengers could actually hear the radio, because the soundproofed Condor was far quieter to travel in than contemporary trimotors. In keeping with the luxury image, the women employed to look after Condor passengers were dubbed “air hostesses,” a term that was to enjoy a long life.

Boeing versus Douglas

But the Condor was essentially a retrograde aircraft. The future lay with all-metal stressed-skin monoplanes reflecting the latest research in aerodynamics and streamlining, and powered by increasingly efficient air-cooled radial engines. In the 1930s the competition for the airliner market turned into a head-to-head between

Douglas and Boeing, both of which produced aircraft that revolutionized expectations of passenger-carrying performance.

Boeing set the ball rolling when it began developing its two-engine Model 247 in 1931. Here was an aircraft that was high-powered and streamlined, and capable of carrying 10 passengers at 155mph (250kph). It also mitigated the rigors of flight with a soundproofed cabin to cut down on engine noise – already reduced by doing without the third motor – and well-upholstered seats to reduce vibration.

As Boeing was tied to United Air Lines, it meant initially to keep the aircraft exclusively for United's use. This provoked TWA vice-president Jack Frye into asking the Douglas company to tender for a rival to the 247. The result was the Douglas Commercial DC-1. In February 1934, in a highly publicized stunt designed as a protest against Roosevelt's decision to transfer airmail to the Army, Frye flew the prototype DC-1 from Burbank, California, to Newark, New Jersey, in 13 hours, despite running into a snowstorm. The model that went into production was the slightly longer DC-2. Like the 247, the DC-2 was a sleek and powerful two-engined all-metal monoplane with the latest features such as NACA engine cowlings and retractable undercarriage. But the DC-2 was faster, had longer range, and, crucially, could carry 14 passengers, offering its operators a potential 40 percent extra revenue compared with the 247. Introduced in 1933, the 247 cut scheduled journey times from coast to coast from 27 hours in the old trimotors to 20 hours, with six refueling stops in place of 14. In service a year later, the DC-2 shaved another two hours off the journey time and cut the number of intermediate stops down to three.

The holy grail

The obvious time for a busy man to travel 18 hours from New York to Los Angeles was overnight, but the DC-2 was not spacious enough to function comfortably as a sleeper. American Airlines' chief C.R. Smith was convinced that an airplane capable of carrying 14 passengers in bunks or 21

UNPROFITABLE SLEEPER

American airline companies wanted to offer sleeper accommodation on coast-to-coast flights to compete with the railroad's Pullman cars. These bunks are on a Douglas Sleeper Transport (DST). Introduced in 1935, the DST could not operate profitably because it carried only 14 passengers. The 21-seat DC-3 (opposite), the “day coach” version of the DST, did make a profit.

**FIRST MODERN AIRLINER**

The 10-passenger Boeing 247 was the first air transport plane to reflect the progress made in engines and streamlining in the early 1930s. It combined a sleek, all-metal, cantilever wing design with retractable landing gear and pneumatic deicing. But the 247 was not a commercial success. Introduced in 1933, it was upstaged by the DC-2 the following year.